

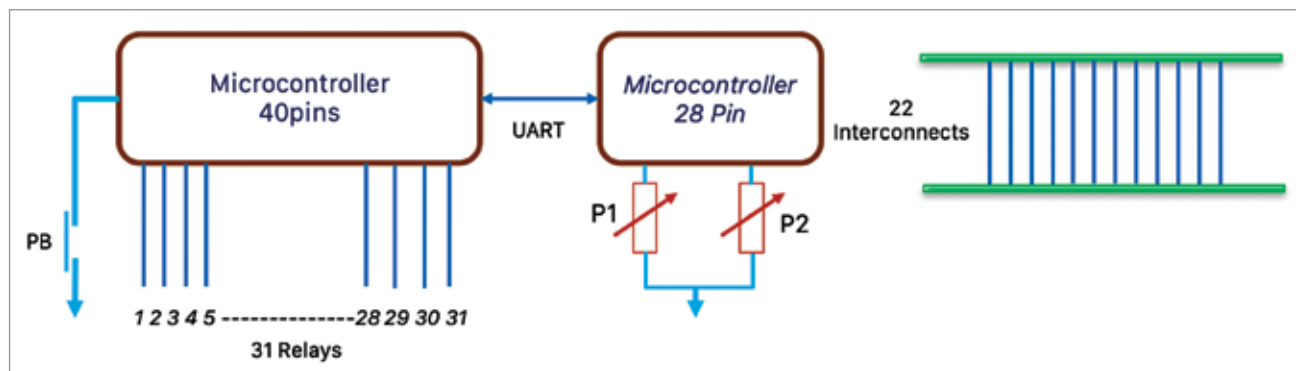


Fig. 1: The original design for a defense application

lowed by good circuit design.

But for this you need experience and depth of knowledge to design the architecture before starting the implementation of the design. What I have observed in the recent times is that, once the team freezes the decision to design a product, they start off with the circuit design, which leads to issues. Some of the critical issues I have seen are:

1. Low performance of the product against requirement
2. Increased cost of product
3. Issues when working in different environments
4. Difficulty in product upgrades

Designing architecture of the product first helps in a robust product. When you do an architecture design, you should not worry about what components you would use, and you should worry about only the functionality of the product. Once architecture is done, implementing the design becomes easier.

Second riskier tendency is getting fixated on 'technology' and customise the design for using the technology. Many a times I have heard startup founders telling me, "Sir, the microcontroller is good, so we are designing a product based on that." But this fallacy leads only to trouble later. As the saying goes, you should buy the right size shoes and not cut the leg for the shoe you like!

The architecture becomes most critical when you design medical devices, avionics, and industrial devices, all of which have safety as the main criterion. Since these are regulated industries (a central agency will approve your product finally), architecture design and fail-safe analy-

sis is critical as review of these designs are a must.

Most designers do not understand the benefits of developing an architecture before they start the design. Advantages of designing the architecture first helps in a product with the following:

1. Predictable performance meeting the needs
2. Simpler design to manufacture and low cost
3. Easy to support and upgrade in future

By the way, designing an architecture is not a rocket science. It just needs basic idea of decomposing a product functions into individual parts and create individual blocks to address those functions.

Why architecture is critical

A well architected product is robust and performs as per the specifications. This effectively results in a successful product with a long life. Remember, for a product to have a long life (medical devices, avionics, and industrial devices), following are important:

1. Performance throughout their long life
2. System should have resources to meet future requirements
3. Reliable and can be upgraded

These requirements call for a well-designed architecture. Long-life products come with their own issues, especially obsolescence. Today seven to ten years is the average life of a semiconductor. So, if the product has to live longer, the design needs to specifically address this challenge.

Second aspect is the cost of the product. As the product ages, customers expect the cost to go down. Added

to this is the fear of obsolescence, which reduces the price of the product continuously. A well architected product results in reduced cost of production and easy maintenance.

In addition to these, product's software development is impacted by the architecture. Well-defined architecture helps in simpler and robust software. This is helpful when the software does not use any RTOS and is developed as a bare metal software. In bare metal software, dependence on the hardware is very high and a well-designed architecture makes the bare metal code writing easy and compact.

A good architecture example

To explain the importance of a good architecture, here is a real-life example of a product which was designed without a good architecture, and the circuit was implemented without going through proper design cycle. This design was created for defense application to replace an existing mechanical control into an electronic control. Main function of the system was to close a series of 31 relays one by one, starting from the first relay. The closure operation started with the press of a push button. The potentiometers P1 and P2 were used to set the following functions:

1. P1 to set the delay time after which the closure of relays would start after pressing of the push button.
2. P2 to set time between closures of relays after the time set by P1, starting from relay 1 to relay 31.

The original design is given in Fig. 1. You can see there are two microcontrollers, and the system has two PCBs due to retrofitting requirement in the existing enclosure. This resulted in 22 interconnect pins between